

Product Sets

If A and B are two nonempty sets, we define the **product set** or **Cartesian product** $A \times B$ as the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$. Thus

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Example 1. Let

$$A = \{1, 2, 3\} \text{ and } B = \{r, s\},$$

then

$$A \times B = \{(1, r), (1, s), (2, r), (2, s), (3, r), (3, s)\}.$$

Observe that the elements of $A \times B$ can be arranged in a convenient tabular array as shown in Figure 4.1. ♦

$\begin{array}{c} B \\ \diagdown \\ A \end{array}$	r	s
1	$(1, r)$	$(1, s)$
2	$(2, r)$	$(2, s)$
3	$(3, r)$	$(3, s)$

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Partitions

A **partition** or **quotient set** of a nonempty set A is a collection $\mathcal{P}$ of nonempty subsets of A such that

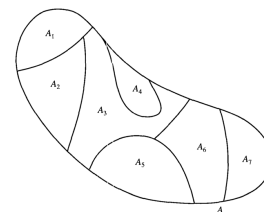
1. Each element of A belongs to one of the sets in $\mathcal{P}$.
2. If A_1 and A_2 are distinct elements of $\mathcal{P}$, then $A_1 \cap A_2 = \emptyset$.

Consider the following subsets of A :

$$A_1 = \{a, b, c, d\}, \quad A_2 = \{a, c, e, f, g, h\}, \quad A_3 = \{a, c, e, g\},$$

$$A_4 = \{b, d\}, \quad A_5 = \{f, h\}.$$

Then $\{A_1, A_2\}$ is not a partition since $A_1 \cap A_2 \neq \emptyset$. Also, $\{A_1, A_5\}$ is not a partition since $e \in A_1$ and $e \in A_5$. The collection $\mathcal{P} = \{A_3, A_4, A_5\}$ is a partition of A . ♦



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Practice

A car manufacturer makes three different types of car frames and two types of engines.

Frame type: sedan (s), coupe (c), van (v)

Engine type: gas (g), diesel (d)

List all possible models of cars.

List all partitions of $B = \{a, b, c, d\}$.

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Relations

Let A and B be nonempty sets. A **relation R from A to B** is a subset of $A \times B$. If $R \subseteq A \times B$ and $(a, b) \in R$, we say that **a is related to b by R** , and we also write $a R b$. If a is not related to b by R , we write $a \not R b$. Frequently, A and B are equal. In this case, we often say that $R \subseteq A \times A$ **is a relation on A** , instead of a relation from A to A .

Example 2. Let A and B be sets of real numbers. We define the following relation R (equals) from A to B :

$$a R b \quad \text{if and only if} \quad a = b. \quad \blacklozenge$$

Example 3. Let

$$A = \{1, 2, 3, 4, 5\}.$$

Define the following relation R (less than) on A :

$$a R b \quad \text{if and only if} \quad a < b.$$

Then

$$R = \{(1, 2), (1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}. \quad \blacklozenge$$

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Example 9. An airline services the five cities c_1, c_2, c_3, c_4 , and c_5 . Table 4.1 gives the cost (in dollars) of going from c_i to c_j . Thus the cost of going from c_1 to c_3 is \$100, while the cost of going from c_4 to c_2 is \$200.

Table 4.1

From \ To	c_1	c_2	c_3	c_4	c_5
c_1		140	100	150	200
c_2	190		200	160	220
c_3	110	180		190	250
c_4	190	200	120		150
c_5	200	100	200	150	

We now define the following relation R on the set of cities $A = \{c_1, c_2, c_3, c_4, c_5\}$:

$c_i R c_j$ if and only if the cost of going from c_i to c_j is less than or equal to \$180.

Find R .

Solution: The relation R is the subset of $A \times A$ consisting of all cities (c_i, c_j) , where the cost of going from c_i to c_j is less than or equal to \$180. Hence

$$R = \{(c_1, c_2), (c_1, c_3), (c_1, c_4), (c_2, c_4), (c_3, c_1), (c_3, c_2), (c_4, c_3), (c_4, c_5), (c_5, c_2), (c_5, c_4)\}. \blacklozenge$$

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Set arising from relations

If R is a relation from A to B and $x \in A$, we define $R(x)$, the **R -relative set of x** , to be the set of all y in B with the property that x is R -related to y . Thus, in symbols,

$$R(x) = \{y \in B \mid x R y\}.$$

Similarly, if $A_1 \subseteq A$, then $R(A_1)$, the **R -relative set of A_1** , is the set of all y in B with the property that x is R -related to y for some x in A_1 . That is,

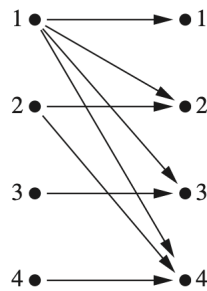
$$R(A_1) = \{y \in B \mid x R y \text{ for some } x \text{ in } A_1\}.$$

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Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

Solution: Because (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b , we see that

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}.$$



R	1	2	3	4
1	×	×	×	×
2		×		×
3			×	
4				×

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Properties of Relations

A relation R on a set A is called **reflexive** if $(a, a) \in R$ for every element $a \in A$.

Consider the following relations on $\{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\},$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$$

$$R_6 = \{(3, 4)\}.$$

Which of these relations are reflexive?

Solution: The relations R_3 and R_5 are reflexive because they both contain all pairs of the form (a, a) , namely, $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$. The other relations are not reflexive because they do not contain all of these ordered pairs. In particular, R_1 , R_2 , R_4 , and R_6 are not reflexive because $(3, 3)$ is not in any of these relations. ▶

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Properties of Relations

- A relation R on a set A is called **symmetric** if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$. A relation R on a set A such that for all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called **antisymmetric**.

Solution: The relations R_2 and R_3 are symmetric, because in each case (b, a) belongs to the relation whenever (a, b) does. For R_2 , the only thing to check is that both $(2, 1)$ and $(1, 2)$ are in the relation. For R_3 , it is necessary to check that both $(1, 2)$ and $(2, 1)$ belong to the relation, and $(1, 4)$ and $(4, 1)$ belong to the relation. The reader should verify that none of the other relations is symmetric. This is done by finding a pair (a, b) such that it is in the relation but (b, a) is not.

R_4 , R_5 , and R_6 are all antisymmetric. For each of these relations there is no pair of elements a and b with $a \neq b$ such that both (a, b) and (b, a) belong to the relation. The reader should verify that none of the other relations is antisymmetric. This is done by finding a pair (a, b) with $a \neq b$ such that (a, b) and (b, a) are both in the relation. 

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Properties of Relations

A relation R on a set A is called **transitive** if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

Solution: R_4 , R_5 , and R_6 are transitive. For each of these relations, we can show that it is transitive by verifying that if (a, b) and (b, c) belong to this relation, then (a, c) also does. For instance, R_4 is transitive, because $(3, 2)$ and $(2, 1)$, $(4, 2)$ and $(2, 1)$, $(4, 3)$ and $(3, 1)$, and $(4, 3)$ and $(3, 2)$ are the only such sets of pairs, and $(3, 1)$, $(4, 1)$, and $(4, 2)$ belong to R_4 . The reader should verify that R_5 and R_6 are transitive.

R_1 is not transitive because $(3, 4)$ and $(4, 1)$ belong to R_1 , but $(3, 1)$ does not. R_2 is not transitive because $(2, 1)$ and $(1, 2)$ belong to R_2 , but $(2, 2)$ does not. R_3 is not transitive because $(4, 1)$ and $(1, 2)$ belong to R_3 , but $(4, 2)$ does not. 

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Combining Relations

Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$. The relations $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ and $R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$ can be combined to obtain

$$R_1 \cup R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\},$$

$$R_1 \cap R_2 = \{(1, 1)\},$$

$$R_1 - R_2 = \{(2, 2), (3, 3)\},$$

$$R_2 - R_1 = \{(1, 2), (1, 3), (1, 4)\}.$$



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Composite Relations

- Let R be a relation from a set A to a set B and S a relation from B to a set C . The *composite* of R and S is the relation consisting of ordered pairs (a, c) , where $a \in A$, $c \in C$, and for which there exists an element $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$. We denote the composite of R and S by $S \circ R$.

What is the composite of the relations R and S , where R is the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$?

Solution: $S \circ R$ is constructed using all ordered pairs in R and ordered pairs in S , where the second element of the ordered pair in R agrees with the first element of the ordered pair in S . For example, the ordered pairs $(2, 3)$ in R and $(3, 1)$ in S produce the ordered pair $(2, 1)$ in $S \circ R$. Computing all the ordered pairs in the composite, we find

$$S \circ R = \{(1, 0), (1, 1), (2, 1), (2, 2), (3, 0), (3, 1)\}.$$



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Representation of relations

Representing Relations Using Matrices

- A relation between finite sets can be represented using a zero–one matrix. Suppose that R is a relation from $A = \{a_1, a_2, \dots, a_m\}$ to $B = \{b_1, b_2, \dots, b_n\}$. The relation R can be represented by the matrix $M_R = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R, \\ 0 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

- the zero–one matrix representing R has a 1 as its (i, j) entry when a_i is related to b_j , and a 0 in this position if a_i is not related to b_j .

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Matrix representation of relations

Suppose that $A = \{1, 2, 3\}$ and $B = \{1, 2\}$. Let R be the relation from A to B containing (a, b) if $a \in A$, $b \in B$, and $a > b$. What is the matrix representing R if $a_1 = 1$, $a_2 = 2$, and $a_3 = 3$, and $b_1 = 1$ and $b_2 = 2$?

Solution: Because $R = \{(2, 1), (3, 1), (3, 2)\}$, the matrix for R is

$$M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

The 1s in M_R show that the pairs $(2, 1)$, $(3, 1)$, and $(3, 2)$ belong to R . The 0s show that no other pairs belong to R . 

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Matrix representation of relations

Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4, b_5\}$. Which ordered pairs are in the relation R represented by the matrix

$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} ?$$

Solution: Because R consists of those ordered pairs (a_i, b_j) with $m_{ij} = 1$, it follows that

$$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3), (a_3, b_5)\}.$$



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Matrix representation of relations

- R on A is reflexive if $(a, a) \in R$ whenever $a \in A$. Thus, R is reflexive if and only if $(a_i, a_i) \in R$ for $i = 1, 2, \dots, n$.
- Hence, R is reflexive if and only if $m_{ii} = 1$, for $i = 1, 2, \dots, n$.
- In other words, **R is reflexive if all the elements on the main diagonal of $\mathbf{M}_R$ are equal to 1.**

$$\begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & \ddots & \\ & & & & & 1 & \\ & & & & & & 1 \end{bmatrix}$$

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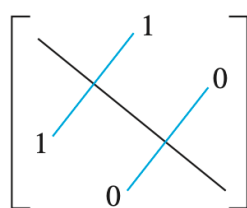
Matrix representation of relations

- The relation R is symmetric if $(a, b) \in R$ implies that $(b, a) \in R$.
Consequently, the relation R on the set $A = \{a_1, a_2, \dots, a_n\}$ is symmetric if and only if $(a_j, a_i) \in R$ whenever $(a_i, a_j) \in R$.
- In terms of the entries of M_R , R is symmetric if and only if $m_{ji} = 1$ whenever $m_{ij} = 1$. This also means $m_{ji} = 0$ whenever $m_{ij} = 0$.
- R is symmetric if and only if

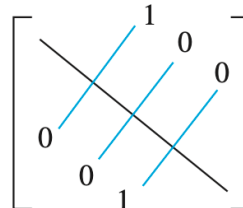
$$M_R = (M_R)^T$$
, that is, if M_R is a symmetric matrix.
- The relation R is antisymmetric if and only if $(a, b) \in R$ and $(b, a) \in R$ imply that $a = b$.
- Consequently, the matrix of an antisymmetric relation has the property that if $m_{ij} = 1$ with $i \neq j$, then $m_{ji} = 0$. Or, in other words, either $m_{ij} = 0$ or $m_{ji} = 0$ when $i \neq j$.

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Matrix representation of relations



(a) Symmetric



(b) Antisymmetric

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Suppose that the relation R on a set is represented by the matrix

$$\mathbf{M}_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Is R reflexive, symmetric, and/or antisymmetric?

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Matrix representation of relations

- The Boolean operations join and meet can be used to find the matrices representing the union and the intersection of two relations.

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2} \quad \text{and} \quad \mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2}.$$

Suppose that the relations R_1 and R_2 on a set A are represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

What are the matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$?

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Matrix representation of relations

- Determining the composite of relations using matrix. Boolean product of the matrices can be used to finding composite of relations.
- suppose that R is a relation from A to B and S is a relation from B to C . Suppose that A , B , and C have m , n , and p elements, respectively. Let the zero-one matrices for $S \circ R$, R and S be $M_{S \circ R} = [t_{ij}]$, $M_R = [r_{ij}]$, and $M_S = [s_{ij}]$, respectively
- The ordered pair (a_i, c_j) belongs to $S \circ R$ if and only if there is an element b_k such that (a_i, b_k) belongs to R and (b_k, c_j) belongs to S . It follows that $t_{ij} = 1$ if and only if $r_{ik} = s_{kj} = 1$ for some k . From the definition of the Boolean product, this means

$$M_{S \circ R} = M_R \odot M_S$$

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Find the matrix representing the relations $S \circ R$, where the matrices representing R and S are

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Solution: The matrix for $S \circ R$ is

$$M_{S \circ R} = M_R \odot M_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

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Find the matrix representing the relation R^2 , where the matrix representing R is

$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

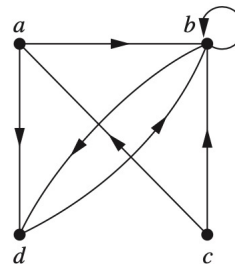
Solution: The matrix for R^2 is

$$\mathbf{M}_{R^2} = \mathbf{M}_R^{[2]} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

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Representing Relations Using Digraphs

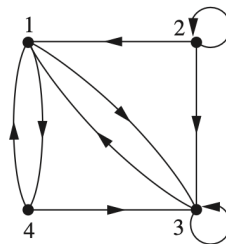
- A directed graph, or digraph, consists of a set V of vertices (or nodes) together with a set E of ordered pairs of elements of V called edges (or arcs). The vertex a is called the initial vertex of the edge (a, b) , and the vertex b is called the terminal vertex of this edge.
- The directed graph with vertices a, b, c , and d , and edges (a, b) , (a, d) , (b, b) , (b, d) , (c, a) , (c, b) , and (d, b) is



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Representing Relations Using Digraphs

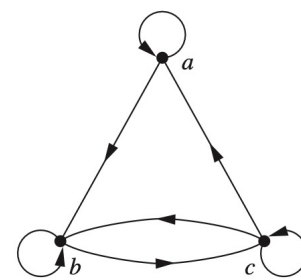
- Represent the given relation into directed graph
 $R = \{(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (4, 1)\}$
 on the set $\{1, 2, 3, 4\}$
- What are the ordered pairs in the relation R represented by the directed graph shown



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Representing Relations Using Digraphs

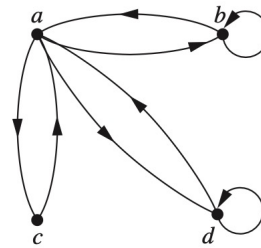
- Determine whether the relations for the directed graphs shown in Figure are reflexive, symmetric, antisymmetric, and/or transitive.
- **Solution:** Because there are loops at every vertex of the directed graph of R , it is reflexive.
- R is neither symmetric nor antisymmetric because there is an edge from a to b but not one from b to a , but there are edges in both directions connecting b and c .
- Finally, R is not transitive because there is an edge from a to b and an edge from b to c , but no edge from a to c .



(a) Directed graph of R

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Representing Relations Using Digraphs



(b) Directed graph of S

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Paths in Directed Graphs

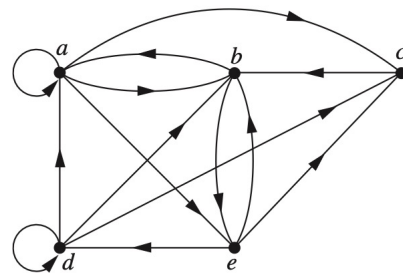
- A path from a to b in the directed graph G is a sequence of edges $(x_0, x_1), (x_1, x_2), (x_2, x_3), \dots, (x_{n-1}, x_n)$ in G , where n is a nonnegative integer, and $x_0 = a$ and $x_n = b$, that is, a sequence of edges where the terminal vertex of an edge is the same as the initial vertex in the next edge in the path.
- This path is denoted by $x_0, x_1, x_2, \dots, x_{n-1}, x_n$ and has length n .
- We view the empty set of edges as a path of length zero from a to a .
- A path of length $n \geq 1$ that begins and ends at the same vertex is called a circuit or cycle.

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Paths in Directed Graphs

- Which of the following are paths in the directed graph shown:

- a, b, e, d;
- a, e, c, d, b;
- b, a, c, b, a, a, b;
- d, c;
- c, b, a;
- e, b, a, b, a, b, e?



- What are the lengths of those that are paths
- Which of the paths in this list are circuits?

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Closure of relations

- let R be a relation on a set A . R may or may not have some property P , such as reflexivity, symmetry, or transitivity. If there is a relation S with property P containing R such that S is a subset of every relation with property P containing R , then S is called the **closure** of R with respect to P .
- A computer network has data centers in Boston, Chicago, Denver, Detroit, New York, and San Diego. There are direct, one-way telephone lines from Boston to Chicago, from Boston to Detroit, from Chicago to Detroit, from Detroit to Denver, and from New York to San Diego.
- Let R be the relation containing (a, b) if there is a telephone line from the data center in a to that in b . How can we determine if there is some (possibly indirect) link composed of one or more telephone lines from one center to another? Because not all links are direct, such as the link from Boston to Denver that goes through Detroit, R cannot be used directly to answer this.
- In the language of relations, R is not transitive, so it does not contain all the pairs that can be linked.
- we can find all pairs of data centers that have a link by constructing a transitive relation S containing R such that S is a subset of every transitive relation containing R . Here, S is the smallest transitive relation that contains R . This relation is called the transitive closure of R .

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Reflective Closure

- The relation $R = \{(1, 1), (1, 2), (2, 1), (3, 2)\}$ on the set $A = \{1, 2, 3\}$ is not reflexive. How can we produce a reflexive relation containing R that is as small as possible?
- This can be done by adding $(2, 2)$ and $(3, 3)$ to R , because these are the only pairs of the form (a, a) that are not in R .
- Clearly, this new relation contains R . Furthermore, any reflexive relation that contains R must also contain $(2, 2)$ and $(3, 3)$. Because this relation contains R , is reflexive, and is contained within every reflexive relation that contains R , it is called the reflexive closure of R .

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Reflective Closure

- given a relation R on a set A , the reflexive closure of R can be formed by adding to R all pairs of the form (a, a) with $a \in A$, not already in R .
- The addition of these pairs produces a new relation that is reflexive, contains R , and is contained within any reflexive relation containing R .
- We see that the reflexive closure of R equals $R \cup \Delta$ where $\Delta = \{(a, a) \mid a \in A\}$ is the diagonal relation on A .

What is the reflexive closure of the relation $R = \{(a, b) \mid a < b\}$ on the set of integers?

Solution: The reflexive closure of R is

$$R \cup \Delta = \{(a, b) \mid a < b\} \cup \{(a, a) \mid a \in \mathbf{Z}\} = \{(a, b) \mid a \leq b\}.$$

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Symmetric Closure

- The relation $\{(1, 1), (1, 2), (2, 2), (2, 3), (3, 1), (3, 2)\}$ on $\{1, 2, 3\}$ is not symmetric. How can we produce a symmetric relation that is as small as possible and contains R ?
- To do this, we need only add $(2, 1)$ and $(1, 3)$, because these are the only pairs of the form (b, a) with $(a, b) \in R$ that are not in R .
- This new relation is symmetric and contains R .
- Furthermore, any symmetric relation that contains R must contain this new relation, because a symmetric relation that contains R must contain $(2, 1)$ and $(1, 3)$. Consequently, this new relation is called the symmetric closure of R .

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Symmetric Closure

- the symmetric closure of a relation R can be constructed by adding all ordered pairs of the form (b, a) , where (a, b) is in the relation, that are not already present in R .
- Adding these pairs produces a relation that is symmetric, that contains R , and that is contained in any symmetric relation that contains R . The symmetric closure of a relation can be constructed by taking the union of a relation with its inverse that is, $R \cup R^{-1}$ is the symmetric closure of R , where $R^{-1} = \{(b, a) \mid (a, b) \in R\}$.

What is the symmetric closure of the relation $R = \{(a, b) \mid a > b\}$ on the set of positive integers?

Solution: The symmetric closure of R is the relation

$$R \cup R^{-1} = \{(a, b) \mid a > b\} \cup \{(b, a) \mid a > b\} = \{(a, b) \mid a \neq b\}.$$

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Transitive Closure

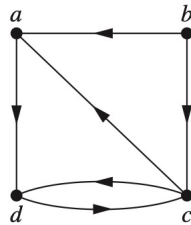
- Suppose that a relation R is not transitive. How can we produce a transitive relation that contains R such that this new relation is contained within any transitive relation that contains R ? Can the transitive closure of a relation R be produced by adding all the pairs of the form (a, c) , where (a, b) and (b, c) are already in the relation?
- Consider the relation $R = \{(1, 3), (1, 4), (2, 1), (3, 2)\}$ on the set $\{1, 2, 3, 4\}$. This relation is not transitive because it does not contain all pairs of the form (a, c) where (a, b) and (b, c) are in R .
- The pairs of this form not in R are $(1, 2)$, $(2, 3)$, $(2, 4)$, and $(3, 1)$. Adding these pairs does not produce a transitive relation, because the resulting relation contains $(3, 1)$ and $(1, 4)$ but does not contain $(3, 4)$.
- This shows that constructing the transitive closure of a relation is more complicated than constructing either the reflexive or symmetric closure.
- The transitive closure of a relation can be found by adding new ordered pairs that must be present and then repeating this process until no new ordered pairs are needed.

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Finding Transitive Closure – Warshall' Algorithm

- Warshall's algorithm is based on the construction of a sequence of zero-one matrices.
- These matrices are $W_0, W_1, \dots, W_n$, where $W_0 = M_R$ is the zero-one matrix of this relation, and $W_k = [w(k)_{ij}]$,
 - where $w(k)_{ij} = 1$ if there is a path from v_i to v_j such that all the interior vertices of this path are in the set $\{v_1, v_2, \dots, v_k\}$ (the first k vertices in the list) and
 - is 0 otherwise. (The first and last vertices in the path may be outside the set of the first k vertices in the list.)
- Note that $W_n = MR^*$, because the (i, j) th entry of MR^* is 1 if and only if there is a path from v_i to v_j , with all interior vertices in the set $\{v_1, v_2, \dots, v_n\}$

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$$W_0 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

$$\{b,c\} \times \{d\} = \{(b,d), (c,d)\}$$

$$W_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

$$\{\Phi\} \times \{a,c,d\} = \{\Phi\}$$

$$W_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

$$\{b,d\} \times \{a,d\} = \{(b,a), (b,d), (d,a), (d,d)\}$$

$$W_3 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}.$$

$$\{a,b,c,d\} \times \{a,c,d\} = \{(a,a), (a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d)\}$$

$$W_4 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}.$$

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Equivalence of Relation

- A relation on a set A is called an equivalence relation if it is reflexive, symmetric, and transitive.
- Let R be the relation on the set of real numbers such that aRb if and only if $a - b$ is an integer. Is R an equivalence relation?

Solution: Because $a - a = 0$ is an integer for all real numbers a , aRa for all real numbers a . Hence, R is reflexive. Now suppose that aRb . Then $a - b$ is an integer, so $b - a$ is also an integer. Hence, bRa . It follows that R is symmetric. If aRb and bRc , then $a - b$ and $b - c$ are integers.

Therefore, $a - c = (a - b) + (b - c)$ is also an integer. Hence, aRc . Thus, R is transitive. Consequently, R is an equivalence relation.

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Equivalence of Relation

- Which of these relations on $\{0, 1, 2, 3\}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack.
 - a) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$
 - b) $\{(0, 0), (0, 2), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$
 - c) $\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$
 - d) $\{(0, 0), (1, 1), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$
 - e) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

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Partial Ordering

- A relation R on a set S is called a partial ordering or partial order if it is reflexive, antisymmetric, and transitive. A set S together with a partial ordering R is called a partially ordered set, or poset, and is denoted by (S, R) . Members of S are called elements of the poset.
- Show that the “greater than or equal” relation $(\geq)$ is a partial ordering on the set of integers.

Solution: Because $a \geq a$ for every integer a , $\geq$ is reflexive. If $a \geq b$ and $b \geq a$, then $a = b$. Hence, $\geq$ is antisymmetric. Finally, $\geq$ is transitive because $a \geq b$ and $b \geq c$ imply that $a \geq c$. It follows that $\geq$ is a partial ordering on the set of integers and $(\mathbb{Z}, \geq)$ is a poset

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Partial Ordering

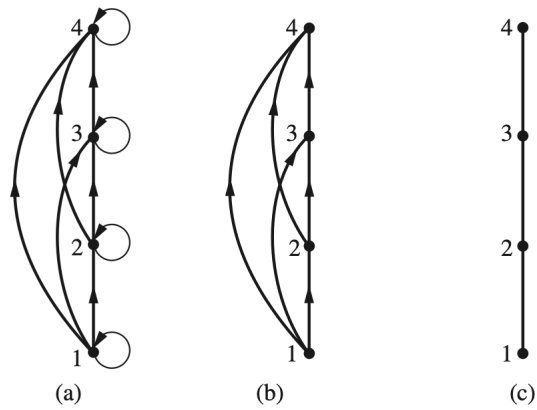
- Which of these relations on $\{0, 1, 2, 3\}$ are partial orderings?
Determine the properties of a partial ordering that the others lack.
 - $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$
 - $\{(0, 0), (1, 1), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$
 - $\{(0, 0), (1, 1), (1, 2), (2, 2), (3, 3)\}$
 - $\{(0, 0), (1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$
 - $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

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Hasse Diagram

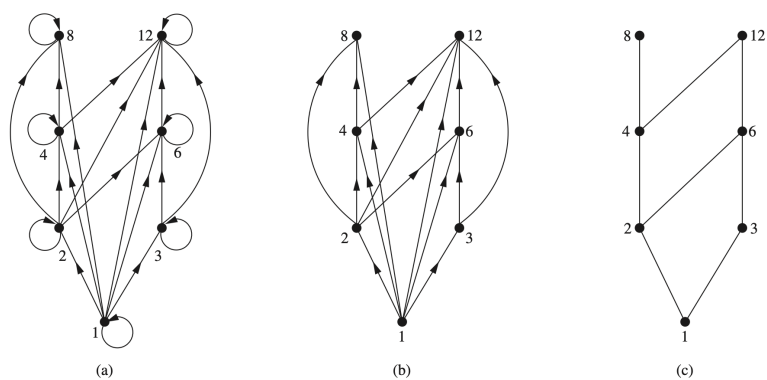
- We can represent a finite poset $(S, <)$ using this procedure:
 - Start with the directed graph for this relation.
 - Because a partial ordering is reflexive, a loop (a, a) is present at every vertex a . Remove these loops.
 - Next, remove all edges that must be in the partial ordering because of the presence of other edges and transitivity. That is, remove all edges (x, y) for which there is an element $z \in S$ such that $x < z$ and $z < y$.
 - Finally, arrange each edge so that its initial vertex is below its terminal vertex.
 - Remove all the arrows on the directed edges, because all edges point “upward” toward their terminal vertex.

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- Draw the Hasse diagram representing the partial ordering $\{(a, b) | a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$.



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